LIFTING AFTER FAITHFUL FLAT DESCENT SUFFICES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1795-1803 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1805-1854, the complete author proof (48 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 26 of queue_stacks-stacks-more-morphisms.json (pb=1986, stmt 1795-1803, proof 1805-1854).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the 1-cocycle verification 'A computation (omitted)' - standard Cech bookkeeping on the displayed mechanism.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07Y9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1854–1854. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07Y6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1854–1854. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1854–1854. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-T-quasi-coherent: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1699–1705. Cited by the

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-sheaf-of-infinitesimal-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1607–1618. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack over a scheme S whose structure morphism $\mathcal{X} \to S$ is locally of finite presentation. Let $(A' \to A, x)$ be a deformation situation. If there exists a faithfully flat finitely presented A'-algebra B' and an object y' of $\mathcal{X}$ over B' lifting $x|_{B' \otimes_{A'} A}$, then there exists an object x' over A' lifting x.*

Proof. Let $I = \mathrm{Ker}(A' \to A)$. Set $B'_1 = B' \otimes_{A'} B'$ and $B'_2 = B' \otimes_{A'} B' \otimes_{A'} B'$. Let $J = IB'$, $J_1 = IB'_1$, $J_2 = IB'_2$ and $B = B'/J$, $B_1 = B'_1/J_1$, $B_2 = B'_2/J_2$. Set $y = x|_B$, $y_1 = x|_{B_1}$, $y_2 = x|_{B_2}$. Let F be the fppf sheaf of Lemma lemma-sheaf-of-infinitesimal-lifts (which applies, see footnote in the proof of Lemma lemma-T-quasi-coherent). Thus we have an equalizer diagram

$$F(A') \longrightarrow F(B') \rightrightarrows F(B'_1)$$

On the other hand, we have $F(B') = \mathrm{Lift}(y, B')$, $F(B'_1) = \mathrm{Lift}(y_1, B'_1)$, and $F(B'_2) = \mathrm{Lift}(y_2, B'_2)$ in the terminology from Artin's Axioms, Section 07Y6. These sets are nonempty and are (canonically) principal homogeneous spaces for $T_y(J)$, $T_{y_1}(J_1)$, $T_{y_2}(J_2)$, see Artin's Axioms, Lemma 07Y9. Thus the difference of the two images of y' in $F(B'_1)$ is an element

$$\delta_1 \in T_{y_1}(J_1) = T_x(I) \otimes_A B_1$$

The equality in the displayed equation holds by Lemma lemma-T-quasi-coherent applied to $A' \to B'_1$ and x and y_1, the flatness of $A' \to B'_1$ which also guarantees that $J_1 = I \otimes_{A'} B'_1$. We have similar equalities for B' and B'_2. A computation (omitted) shows that δ_1 gives a 1-cocycle in the Čech complex

$$T_x(I) \otimes_A B \to T_x(I) \otimes_A B_1 \to T_x(I) \otimes_A B_2 \to \ldots$$

By Descent, Lemma 03FJ this complex is acyclic in positive degrees and has $H^0 = T_x(I)$. Thus we may choose an element in $T_x(I) \otimes_A B = T_y(J)$ whose boundary is δ_1. Replacing y' by the result of this element acting on it, we find a new choice y' with $\delta_1 = 0$. Thus y' maps to the same element under the two maps $F(B') \to F(B'_1)$ and we obtain an element o $F(A')$ by the sheaf condition. □